

CURRICULUM VITAE

Braden J. Marazzo-Nowicki

Notarization: I have read the following and certify that this *curriculum vitae* is a current and accurate statement of my professional record.

Signature _____

Date October 1, 2026

I. Personal Information

I.A Contact Information

Braden J. Nowicki (or Marazzo-Nowicki) [They/Them]
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I.B Educational Background

August 2026—Present *McGill University*, Montréal, QC, Canada

Anticipated: Master of Science (MSc) 2028 and Doctor of Philosophy (PhD) 2032, Physics
Advisor: Professor of Physics Daryl Haggard, <https://www.dhaggard.physics.mcgill.ca/>

August 2022—May 2026 *University of Maryland*, College Park, MD, US

Bachelor of Science (BSc), Astronomy with Minor in Computer Science (May 20, 2026)
GPA: 3.991, Major GPA: 4.00
Astronomy Departmental High Honours (for Undergraduate Thesis with Dr. Massimo Ricotti)

September 2025—December 2025 *University of Edinburgh*, Edinburgh, Scotland, UK
Visiting Student, Fall 2025

I.C Professional Societies

2024—Present American Astronomical Society (AAS)

II. Research and Scholarly Activities

II.A Current Research Interests

Gravitational Astrophysics — Black Hole Imaging — Photon Ring Science — Accretion and Plasma Physics
— Active Galactic Nuclei — Jet Formation — Primordial/High Redshift Galaxies — Cosmology — Radio
Astronomy — Very Long Baseline Interferometry

II.B Publications

1. Marazzo-Nowicki, B. J., Tiede, P., Chang, D. O., Palumbo, D. C. M., & Johnson, M. D., The Astrophysical Journal, 2026, <https://doi.org/10.3847/1538-4357/ae9d68>, arXiv: arxiv.org/abs/2601.16267 (Inferring the Black Hole and Accretion Parameters of Sagittarius A* from EHT Observations Using a Semianalytic Model)

II.C Presentations/Conference Proceedings

1. Nowicki, B., & Ricotti, M. 2025, in American Astronomical Society Meeting Abstracts, Vol. 245, 250.24, <https://baas.aas.org/pub/2025n2i250p24>
2. Marazzo-Nowicki, B. J., Tiede, P., Chang, D. O., Palumbo, D. C. M., Johnson, M. D. 2026, in American Astronomical Society Meeting Abstracts, Vol. 247, 158.01, NASA ADS
3. Marazzo-Nowicki, B. J., Supermassive Black Holes and Blue Notes

II.D Research Positions

June 2, 2025–January 10, 2026 *Astrophysics Research Intern, Harvard & Smithsonian Center for Astrophysics*, Cambridge, MA, US. Advisors: Dr. Paul Tiede, Dr. Dominic Chang, PI: Dr. Michael Johnson. Collaborations: Black Hole Initiative (BHI), Event Horizon Telescope (EHT), Black Hole Explorer (BHEX), Smithsonian Astrophysical Observatory (SAO) Radio and Geoastrometry Division.

Project: *Measuring the Black Hole and Accretion Parameters of Sagittarius A* from EHT Observations using a Semi-Analytic Model: Bayesian Black Hole Parameter Inference and Photogrammetry*

- Sagittarius A*—the radio source associated with the supermassive black hole (SMBH) at the center of the Milky Way galaxy which was observed by the EHT Collaboration—displays significant time variability, preventing traditional Very Long Baseline Interferometry (VLBI) methods for increasing (u, v) sampling like Earth-rotation aperture synthesis. We perform “snapshot modeling”—Bayesian parameter inference on complex visibilities from short, 120 s snapshots, directly fitting physical parameters of interest along with complex station gains. We fit a semi-analytic, ray-traced dual-cone accretion flow and emission model in a Kerr spacetime. I explore the multimodal, high-dimensional posteriors and uncertainties with parallel-tempered Markov Chain Monte Carlo (PTMCMC) sampling. The snapshot posteriors are stacked in a Bayesian hierarchical model to extract persistent source structure and variability characteristics.
- We find that spin is poorly constrained without additional methodological approaches and modeling assumptions, as expected. The inference favors a nearly face-on observer inclination, an emission peak near the horizon, a nearly vertical projected spin position angle, and significant emission above the equatorial plane. Future inference will employ more model considerations, such as an optical depth prescription, multi-component emission (equatorial disk and additional large-scale diffuse structures), and non-axisymmetric fluid flows.
- The methodology used here will continue to be used and adapted to improve our understanding of the SMBH and its surrounding magnetized plasma accretion flows. The pipeline is available to use on all past and future EHT Sgr A* observational campaigns to tighten parameter constraints and characterize inter-year variability.
- Presented at the Center for Astrophysics, Black Hole Initiative, and Massachusetts Institute of Technology Haystack Observatory.
- *Conference Presentation*: Gave an oral presentation at AAS247 on January 5, 2026.
- *Working Group Presentations*: Presentation of research in the EHT Analysis Working Group, November 26, 2025. BHEX Plenary Presentation, February 4, 2026.
- *Publication*: Paper submitted to the Astrophysical Journal. This will present the first measurements of the Sgr A* observer inclination and other parameters using cutting-edge accretion disk modeling and Bayesian inference methods with a statistical treatment of the variability power spectrum. NASA ADS

January 2024–Present *Undergraduate Astrophysics Researcher, UMD Astronomy Department*, College Park, MD, US. Advisor: Professor Massimo Ricotti.

Project: *Understanding Nebular Emission Diagnostics and Spectra in JWST Observations using Early Galaxy Simulations*. Developed Software Pipeline: Merlin.

- Developed a pipeline for producing spectra and narrowband images from cosmological simulations of early universe/high redshift galaxies. This involves running a suite of photoionization simulations for a large parameter space and integrating with data output from cosmological simulations.
- The pipeline is used to visualize nebular emission and spectra over time with changing physical conditions in simulated galaxies as they evolve, analyse line luminosity responses to star formation bursts and other evolutionary processes, and evaluate the applicability of nebular line ratios as observational indicators/diagnostics of physical conditions in the early universe. In the future, fitting high-redshift galaxy spectra from James Webb Space Telescope (JWST) data may enable physical parameter inference.
- Supporting younger students in the Gemstone Honors Program who use the package as part of Professor Ricotti's research group.
- *Conference Presentation*: Presented research poster at the AAS245 meeting in January, 2025.
- *Departmental Honours Undergraduate Thesis*: After defending this work on April 20, 2026, I received *High Honours in the University of Maryland Department of Astronomy*.

June 3, 2024–August 9, 2024 *Astrophysics Research Intern, NASA Goddard Space Flight Center, Astrophysics Science Division*, Greenbelt, MD, US. Advisors: Dr. Ronald Gamble (NASA GSFC Code 660, University of Maryland College Park), Dr. Leonid Petrov (NASA GSFC Code 61A), and Dr. Scott Noble (NASA GSFC Code 663). Collaborations: Black Hole Explorer (BHEX).

Project: *Simulating Very Long Baseline Interferometry (VLBI) Observations of Low Luminosity Active Galactic Nuclei (LLAGN) and Ray-Traced General Relativistic Magnetohydrodynamics (GRMHD) Black Hole Images*.

- Simulating (in Fortran) Very Long Baseline Interferometry (VLBI) radio observations/visibilities of Active Galactic Nuclei (AGN) and ray-traced General Relativistic Magnetohydrodynamics (GRMHD) black hole images. The visibility simulations are used to understand what features of a black hole and its accretion disk a proposed space-telescope extension to the Event Horizon Telescope (EHT) would be able to reliably resolve, in connection with the Black Hole Explorer (BHEX) Mission.
- To understand the environment around black holes like M87*, we also worked on modelling black hole environments and spacetime structure, such as plotting photon orbits (along null geodesics), considering observational features like the photon ring, and analysing equations related to the Blandford-Znajek jet formation process.
- Presented research poster at NASA Goddard Space Flight Center.

January 2023–December 2023 *Undergraduate Research Assistant, UMD Physics Department*, College Park, MD, US, US. Advisor: Professor Shabnam Jabeen. Partnerships: National QLab at UMD, IonQ.

Project: *Developing Quantum Computer Neural Networks to Classify the Presence of the Higgs Boson in Large Hadron Collider (LHC) Particle Detection Data*.

- Researched the application of Quantum Neural Networks/Machine Learning (using IBM's Qiskit in Python) to Large Hadron Collider (LHC) particle detection data in partnership with *National QLab* and *IonQ*.
- We developed a 93% accurate Quantum Neural Network classifier to determine whether the Higgs Boson particle was present and decayed in a collision based on features of the data obtained by the Compact Muon Solenoid detector. Our work continues to be developed and expanded upon by the research group.
- Presented research poster in the physics department.

II.E Software Packages

1. **Published: 2025** Merlin—Nebular Line Emission Diagnostics from Cosmological Simulations of Early Universe Galaxies.

II.F Honours, Awards/Grants, and Accomplishments

- *University of Toronto Connaught International Scholarship, \$50,000 CAD over five years*, offered and declined, 2026
- *Astronomy Departmental High Honours*
- *Summa Cum Laude Latin Honours*, 2026
- *University of Maryland Honours Research Grant, \$500*, Honours Awards List, Spring 2026.
- *National Science Foundation (NSF) Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support (ACCESS)* allocation PHY250202: Semianalytical snapshot modeling of Sgr A* with the Event Horizon Telescope—400,000 compute credits awarded.
- *Maryland Space Grant, \$2,000*: awarded by the National Space Grant Consortium (facilitated by NASA) to promising undergraduates in space science, 2025–2026
- *UMD Education Abroad Grant, \$1,000*, Fall 2025 Semester
- *UMD Philip Merrill Presidential Scholar*, 2025–2026; Declined (unable to attend an event).
- *UMD College of Computer, Mathematical, and Natural Sciences Dean's List/Academic Honours*, Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024, Spring 2025 Semesters.
- *UMD President's Scholarship*, Fall 2022–Spring 2026 Semesters (All four years of university).

III. Additional Experience

III.A Technical Projects

Spring 2026 *Message Passing Interface (MPI) Parallel Distributed Memory Implementations of Heat Propagation Code and K-Means Clustering*

Spring 2026 *OpenMP and GPU CUDA Parallel Shared Memory Implementations of K-Means Clustering*

Fall 2024 *Investigating the H α and [O III] Intensities of HII Regions in Messier 33 with a Focus on NGC 604*

- Imaged a portion of Messier 33 (the Triangulum Galaxy) including NGC 604 with a UMD Observatory Telescope.
- Wrote aperture photometry and edge detection routines to determine that emission nebulae in M33 typically appear quantifiably larger and brighter in the H α narrowband filter than in [O III] filter.
- Reported work in a paper for an Observational Astronomy course.

Fall 2024 *Analysing the Performance of Artificial Intelligence Algorithms in the Traveling Salesperson Problem*

- Implemented heuristic algorithms (nearest neighbors, nearest neighbors with 2-opt, repeated randomized nearest neighbors with 2-opt), A-Star with a Minimum Spanning Tree heuristic, and local search algorithms (hill-climbing, simulated annealing, genetic algorithm) to solve instances of the Traveling Salesperson Problem.
- Quantitatively analysed and compared the performance and efficiency of each algorithm.

Fall 2024 *Probabilistic Reasoning and Robot Localization*

- Implemented a particle filter, Kalman filter, and Bayesian network to solve vehicle localization problems (given sensor readings as the car moves) and probabilistic reasoning problems.

Spring 2024 *Studying Factors Related to United States Bird Population Decline with Machine Learning*

- Data analysis incorporating multiple large-scale data sets. In a group, we made a machine learning model to characterize the significance of various factors (deforestation, urban index, building density, climate/temperature change) in predicting bird population decline in counties across the United States.

- Wrote a comprehensive report outlining the entire data analysis pipeline (Data Collection, Data Processing, Exploratory Data Analysis and Visualization, Hypothesis Testing and Machine Learning, Insight and Policy Decisions).

Fall 2023 *Lambda Calculus Compiler*

- Developed algorithms for lexing, parsing, and interpreting Lambda Calculus in OCaml.

Fall 2023 *Regular Expression Engine*

- Wrote algorithms to implement and work with Nondeterministic Finite Automata, Deterministic Finite Automata, and regular expressions in python.

Fall 2023 *Confirming the Transit of Exoplanet WASP-52b Around its Host Star*

- Imaged WASP-52 over the course of an expected exoplanet transit with a UMD Observatory Telescope.
- Retrieved a light curve with the characteristic dip (ingress and egress) during transit using differential photometry.

III.B Skills

Programming Languages/Technologies: Python (including libraries like SciPy, Numpy, Pandas, Matplotlib, etc.), C, Julia, Java, MATLAB, Fortran, Assembly, OCaml, Rust, HTML, JavaScript, CSS, React, SQL, Linux, UNIX, Windows, MacOS, Wolfram Mathematica, LaTeX, Shell Scripting (bash), Prolog, Slurm/High Performance Computing (HPC) Clusters, OpenMP, Message Passing Interface (MPI), CUDA GPU Computing, HPC Toolkit, C-BLAS Linear Algebra Library

Additional Skills: Computational and Theoretical Astrophysics, VLBI Imaging, Bayesian Inference, Black Hole Photogrammetry, Sampling Algorithms (Markov Chain Monte Carlo, Parallel Tempering, Nested Sampling, etc.), Numerical Analysis Algorithms, Machine Learning Analysis, Data Science (Scraping, Tidying, Visualization, Analysis), Observational Astronomy, Astronomy Software (AstroImageJ, ASTAP, Siril), Dissemination of Scientific Research (Presentation, Writing), Parallel Computing, Artificial Intelligence (AI) Paradigms, Computational Astrophysics Codes (Cloudy Photoionization Code, yt-project, RAMSES-RT, MESA, astropy)

III.C Relevant Coursework

Physics/Astrophysics, Mathematics:

- Mathematical Methods for Physics I & II (Linear Algebra, Vector Calculus, Fourier Analysis, Ordinary Differential Equations, Partial Differential Equations, Complex Analysis). Textbook: *Mathematical Methods in the Physical Sciences*, Mary L. Boas. [PHYS274 3 credits & PHYS373 3 credits]
- Physics II: Fields (E & M) (previous credit). [PHYS260 3 credits]
- Experimental Physics II: E & M [PHYS276 2 credits]
- Physics III: Waves. Textbook: *Waves*, David Morin. [PHYS273 3 credits]
- Modern Physics. *Modern Physics for Scientists and Engineers, Fourth Edition*, Andrew F. Rex and Stephen Thornton. [PHYS371 3 credits]
- Introductory Astrophysics I & II. [ASTR120 3 credits & ASTR121 4 credits]
- Observational Astronomy. Textbook: *Observational Astronomy, Second Edition*, D. Scott Birney, Guillermo Gonzalez, and David Oesper. [ASTR310 4 credits]
- Calculus III: Multivariable Calculus. Textbook: *Calculus, Sixth Edition*, Robert Ellis and Denny Gulick. [MATH241 4 credits]
- Probability Theory. Textbook: *A First Course in Probability, Tenth Edition*, Sheldon Ross. [STAT410 3 credits]
- Special Problems in Astronomy (Research). [ASTR498 4 credits, 2 semesters]
- Astronomy Departmental Honours Research/Undergraduate Thesis. [ASTR399 4 credits]

- Graduate Computational Astrophysics (Visualization, Numerical Integration and Differentiation Algorithms, Linear and Nonlinear Equations, Regression, Numerical Ordinary and Partial Differential Equation Solving, N-Body Techniques, Fluid Dynamics, Fourier Transforms, Parallel Computing, Deep Learning). Textbook: *Computational Physics*, Mark Newman. [ASTR615 3 credits]
- Quantum Physics I. Textbook: *Quantum Mechanics: A Paradigms Approach*, David H. McIntyre. [PHYS401 4 credits]
- Theoretical Astrophysics. Textbooks: *Principles of Astrophysics*, Charles Keeton; *Astrophysics for Physicists, 2010*, Arnab Rai Choudhuri; *Astrophysics in a Nutshell, 2007*, Dan Maoz. [ASTR320 3 credits]
- Stellar Structure and Evolution. Textbook: *Understanding Stellar Evolution*, Emily Levesque and Henry J. G. L. M. Lamers. [ASTR406 3 credits]
- Graduate Radiation Processes in Astrophysics (at The University of Edinburgh; combines High Energy Astrophysics and Radiative Processes). Textbooks: *High Energy Astrophysics Volumes 1 and 2, Third Edition (Combined)*, Malcolm Longair; *Radiative Processes in Astrophysics*, George B. Rybicki and Alan P. Lightman; *The Physics of Astrophysics, Volume 1: Radiation*, Frank Shu; *An Introduction to Modern Astrophysics*, Bradley W. Carroll and Dale A. Ostlie. [PHYS11067 20 Edinburgh credits, 6 UMD Credits]
- Lagrangian Dynamics (at The University of Edinburgh). Textbooks: *Classical Mechanics*, R. Douglas Gregory. [PHYS10015 10 Edinburgh credits, 3 UMD credits]
- Quantum Physics II. Textbook: *Quantum Mechanics: A Paradigms Approach*, David H. McIntyre. [PHYS402 3 credits]
- Introduction to Statistical Thermodynamics. Textbook: *An Introduction to Thermal Physics*, Daniel V. Schroeder [PHYS404 3 credits]

Computer Science:

- Object Oriented Programming II (Java). [CMSC132 4 credits]
- Introduction to Computer Systems (C, Assembly). [CMSC216 4 credits]
- Honours Algorithms. [CMSC351H 3 credits]
- Organization of Programming Languages (Python, Rust, OCaml). [CMSC330 3 credits]
- Discrete Structures. Textbook: *Discrete Mathematics*, Susanna Epp. [CMSC250 4 credits]
- Data Science (Python). [CMSC320 3 credits]
- Introduction to Artificial Intelligence (Python, Prolog). Textbook: *Artificial Intelligence: A Modern Approach, Fourth Edition 2020*, Stuart Russell and Peter Norvig. [CMSC421 3 credits]
- Introduction to Parallel Computing. Textbook: *Parallel Programming for Science and Engineering*, Victor Eijkhout. [CMSC416 3 credits]

III.D Volunteer/Service Work

September 2026—Present *Equity, Diversity, and Inclusion (EDI) Officer, McGill Graduate Association of Physics Students (MGAPS)*

September 2025—Present *Member, Event Horizon Telescope (EHT) Responsible Siting Committee*

- Working with anthropologists, representatives, and local communities to ensure responsible, ethical siting and practices for telescopes/observatories in the EHT network.
- Developing a responsible siting framework/paper in advance of the next Astronomy Decadal Survey to disseminate across the field and promote ethical siting practices. I am primarily contributing discussion of epistemic violence and the importance of considering historically excluded ontologies/ways of knowing.

January 2026—May 2026 *Member/Organiser, 17 For Peace and Justice*, College Park, MD, US

- Organiser in the community action subcommittee, working to organise events and outreach addressing environmental injustice locally and globally.

September 2025—December 2025 *Volunteer, Dirty Weekenders Society*, Edinburgh, Scotland, UK

- Weekly conservation work in the local community.

September 2025—December 2025 *Volunteer*, **Permaculture Garden Society**, Edinburgh, Scotland, UK

- Work in a community garden at the University of Edinburgh.

June 2025—August 2025 *Volunteer*, **Youth to Youth Homeless Shelter**, Cambridge, MA, US

- Weekly work in the Youth to Youth Homeless Shelter in Cambridge, providing services for youths experiencing homelessness.

January 2025—May 2025 *Astronomy Tutor*, **UMD Astronomy Department**, College Park, MD, US

- Supporting students in introductory astrophysics courses with homework, labs, and conceptual understanding. The tutoring program was later discontinued due to funding availability.

February 2025—Present *Volunteer*, **Homes Not Borders**, DC Area

- Furnishing and setting up apartments for refugee families in the local area.

April 2024—Present *Volunteer Water Quality Monitor*, **Anacostia Riverkeepers**

- Water quality monitoring, clean ups, and advocacy for the Anacostia Watershed in the DC-Maryland-Virginia Area. Working to repair decades of pollution and abuse.

September 2023—December 2023 *Programmer*, **UMD Hack4Impact**, College Park, MD, US

- Applied web development techniques and technologies (HTML, JavaScript, CSS, React) to specific requests from partnered nonprofits.

July 2021—August 2022 *Volunteer*, **Habitat for Humanity Restore**, Lawrence MA, US

- Provided low-cost house materials for low-income families.
- Handled stock/donations, organized, and cleaned.